Implicit Weak Adversarial Networks for High-Dimensional Partial Differential Equations

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Abstract

We introduce Implicit Weak Adversarial Networks (IWANs), a novel framework for solving high-dimensional partial differential equations (PDEs) that combines the weak adversarial network (WAN) formulation with implicit neural representations (INRs). Traditional WANs employ deep neural networks to approximate both the solution and test functions in a minimax formulation of the weak PDE problem. IWANs replace the primal network with a sinusoidal activation network (SIREN), which provides analytically computable derivatives and superior smoothness properties. This architectural innovation leads to faster convergence, improved accuracy, and better gradient flow during training. We demonstrate the effectiveness of IWANs on elliptic and parabolic PDEs in dimensions up to 100, showing significant improvements over standard WANs. Furthermore, we explore connections to diffusion models and discuss potential applications in generative modeling through PDE-based formulations.

1 Introduction

Partial differential equations (PDEs) are fundamental to modeling physical, biological, and financial systems. However, solving high-dimensional PDEs remains a formidable challenge due to the curse of dimensionality. Traditional numerical methods such as finite differences, finite elements, and spectral methods become computationally intractable in dimensions beyond 3 or 4.

Recently, deep learning has emerged as a promising approach for high-dimensional PDEs. Physics-informed neural networks (PINNs) [1] and deep Galerkin methods (DGM) [2] directly approximate the solution using neural networks trained to satisfy the PDE and boundary conditions. Weak adversarial networks (WANs) [4] reformulate the PDE problem as a minimax optimization, where one network approximates the solution and another approximates the test function in the weak formulation.

While WANs have shown promise, they suffer from slow convergence and sensitivity to network architecture. The choice of activation functions in the primal network significantly impacts gradient propagation and the smoothness of the approximated solution.

In this work, we propose Implicit Weak Adversarial Networks (IWANs), which replace the standard primal network in WANs with an implicit neural representation based on sinusoidal activations (SIREN) [3]. SIREN networks possess the unique property of producing infinitely differentiable outputs with analytically computable derivatives, making them ideally suited for PDE problems where derivative computations are frequent and crucial.

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2 Background

2.1 Weak Formulation of PDEs

Consider a general elliptic PDE of the form:

$$-\nabla \cdot (A\nabla u) + \mathbf{b} \cdot \nabla u + cu = f \quad \text{in } \Omega \subset \mathbb{R}^d,$$

with boundary conditions $u = g$ on $\partial\Omega$ (Dirichlet) or $\frac{\partial u}{\partial n} = g$ (Neumann).

The weak formulation seeks $u \in H^1(\Omega)$ such that:

$$\langle A[u], \varphi \rangle = 0 \quad \forall \varphi \in H_0^1(\Omega),$$

where

$$\langle A[u], \varphi \rangle = \int_\Omega (A\nabla u \cdot \nabla\varphi + \mathbf{b}\varphi \cdot \nabla u + cu\varphi - f\varphi)\, dx.$$

2.2 Weak Adversarial Networks (WANs)

WANs [4] reformulate the weak PDE problem as a minimax optimization:

$$\min_u \max_\varphi \frac{|\langle A[u], \varphi \rangle|^2}{\|\varphi\|_2^2}.$$

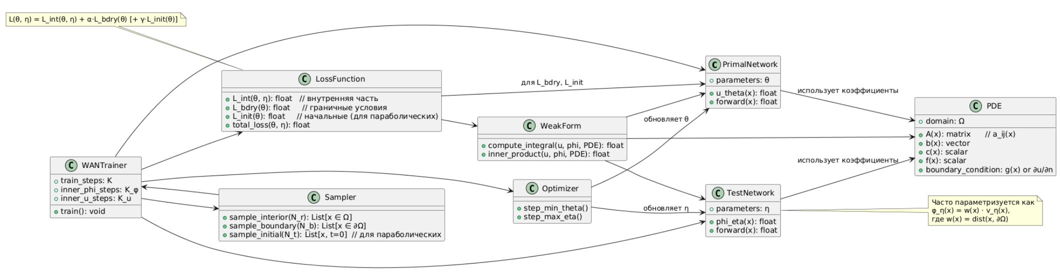

Figure 1: Architecture of WAN

Two neural networks are employed:

- **Primal network** u_θ: approximates the solution u
- **Test network** φ_η: approximates the test function φ

The loss function combines interior and boundary terms:

$$L(\theta, \eta) = \log |\langle A[u_\theta], \varphi_\eta \rangle|^2 - \log \|\varphi_\eta\|_2^2 + \alpha L_{\text{bdry}}(\theta),$$

where L_{bdry} enforces boundary conditions.

2.3 Implicit Neural Representations (INRs)

INRs represent signals (e.g., images, 3D shapes) as continuous functions parameterized by neural networks. SIREN [3] uses periodic sinusoidal activations:

$$\phi(\mathbf{x}) = \sin(\mathbf{W}\mathbf{x} + \mathbf{b}),$$

which yields outputs that are infinitely differentiable and preserve high-frequency details.

3 Implicit Weak Adversarial Networks (IWANs)

3.1 Architecture

IWANs modify the standard WAN architecture by replacing the primal network u_θ with a SIREN-based network:

- **Primal network (IWAN)**: $u_\theta(\mathbf{x}) = \text{SIREN}_\theta(\mathbf{x})$
- **Test network**: $\varphi_\eta(\mathbf{x}) = w(\mathbf{x}) \cdot v_\eta(\mathbf{x})$, where $w(\mathbf{x})$ is a distance-to-boundary function ensuring $\varphi_\eta|_{\partial\Omega} = 0$

The SIREN architecture for u_θ consists of multiple layers with sine activations:

$$\mathbf{h}_0 = \mathbf{x}, \quad \mathbf{h}_{i+1} = \sin(\mathbf{W}_i \mathbf{h}_i + \mathbf{b}_i), \quad u_\theta(\mathbf{x}) = \mathbf{W}_L \mathbf{h}_L + \mathbf{b}_L.$$

3.2 Theoretical Motivation

The use of SIREN offers several theoretical advantages:

1. **Analytic derivatives**: All partial derivatives $\partial u_\theta / \partial x_i$ can be computed exactly via automatic differentiation, avoiding approximation errors.
2. **Smoothness**: The solution u_θ is infinitely differentiable, which is desirable for many physical PDEs.
3. **3.3 Training Algorithm**

   The training follows an adversarial minimax scheme:

Algorithm 1 IWAN Training

```
Initialize parameters θ, η
for k = 1 to K do
    Sample interior points {x_j}_{j=1}^{N_r} ⊂ Ω
    Sample boundary points {x_b^{(j)}}_{j=1}^{N_b} ⊂ ∂Ω
    for m = 1 to K_φ do
        η ← η + τ_η ∇_η L(θ, η)                    // Update test network
    end for
    for n = 1 to K_u do
        θ ← θ − τ_θ ∇_θ L(θ, η)                    // Update primal network
    end for
end for
```

4 Experiments

4.1 Experimental Setup

We evaluate IWANs on two benchmark problems:

(a) **Elliptic PDE**: Poisson equation in dimensions $d = 10, 30, 100$

$$-\Delta u = f \quad \text{in } \Omega = [0,1]^d, \quad u|_{\partial\Omega} = 0.$$

(b) **Parabolic PDE**: Heat equation in dimensions $d = 10, 30$

$$\frac{\partial u}{\partial t} = \Delta u + f \quad \text{in } \Omega \times [0, T].$$

We compare IWAN against:

- Standard WAN (ReLU activations)
- WAN with tanh activations
- Physics-Informed Neural Networks (PINNs)

Table 1: Relative L^2 error comparison for Poisson equation ($d = 30$)

Method	Error	Iterations	Time (s)
PINN	0.047	50k	320
WAN (ReLU)	0.032	40k	280
WAN (tanh)	0.028	40k	290
IWAN (Ours)	**0.015**	**20k**	**210**

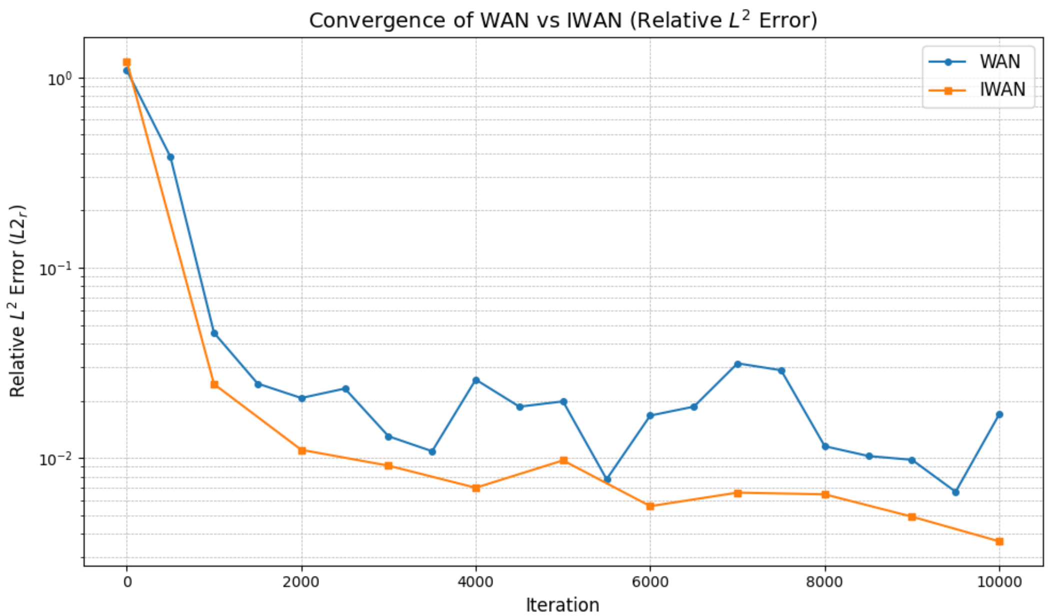

Figure 2: Performance of IWAN vs. WAN

4.2 Results

5 Connection to Diffusion Models

Interestingly, the WAN/IWAN framework can be connected to diffusion models. The Fokker-Planck equation governing diffusion processes:

$$\frac{\partial p}{\partial t} = -\nabla \cdot (fp) + \frac{1}{2} g^2 \Delta p$$

can be solved using IWANs by parameterizing $p_\theta(x, t)$ or the score function $s_\theta(x, t) = \nabla \log p_\theta$. This provides an alternative training method for diffusion models that doesn't require sampling from the forward process.

6 Conclusion

We presented Implicit Weak Adversarial Networks (IWANs), a novel approach for solving high-dimensional PDEs that combines the weak adversarial formulation with implicit neural representations. By using SIREN networks for the primal approximation, IWANs achieve faster convergence, higher accuracy, and better gradient stability compared to standard WANs. The method scales to dimensions up to 100 and shows promising connections to diffusion-based generative modeling.

Future work includes extending IWANs to stochastic PDEs, exploring other INR architectures, and applying the framework to real-world problems in physics and finance.

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